



CPSC 100

Computational Thinking

Generative Artificial Intelligence

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Agenda

- About Me
- Generative AI
 - Background
 - Large Language Model (LLM)
- Co-creating a CPSC 100 AI policy
 - Class Activity

Learning Goals



Learning Goals

After this lecture, you should be able to:

- Explain the concept of **Generative Artificial Intelligence**
 - Describe the relevance of Generative AI in CPSC 100
- Identify and list various GenAI tools to date
- Identify opportunities to use and not use GenAI
- Develop a classroom AI policy

Generative AI

Generative AI

What does Generative mean?

Generative refers to systems, models, or algorithms designed to **create or produce new data that resembles the input data** on which they were trained.

Generative AI

What does AI mean?

Artificial Intelligence (AI), which refers to the **simulation of human intelligence** in machines that are programmed to think, reason, learn, and **act in a way** that mimics human cognitive abilities

Background Information

Hello ChatGPT!

- Previous conversational chatbots (~1960-2000s)
- ELIZA^[1], PARRY^[2], A.L.I.C.E.^[3] & Cleverbot ^[4]
- Rise of ChatGPT-3.5 in November 2022
 - Chat **G**enerative **P**re-trained **T**ransformer
- Wide range of capabilities
 - Human-like responses ^[5]
 - Understanding conversation context ^[6]
 - Writing code & debugging ^[7]



Chatbots are NOT new!

Unleashing the Potential of Chatbots in Education: A State-Of-The-Art Analysis

Rainer Winkler and Matthias Soellner March 2018

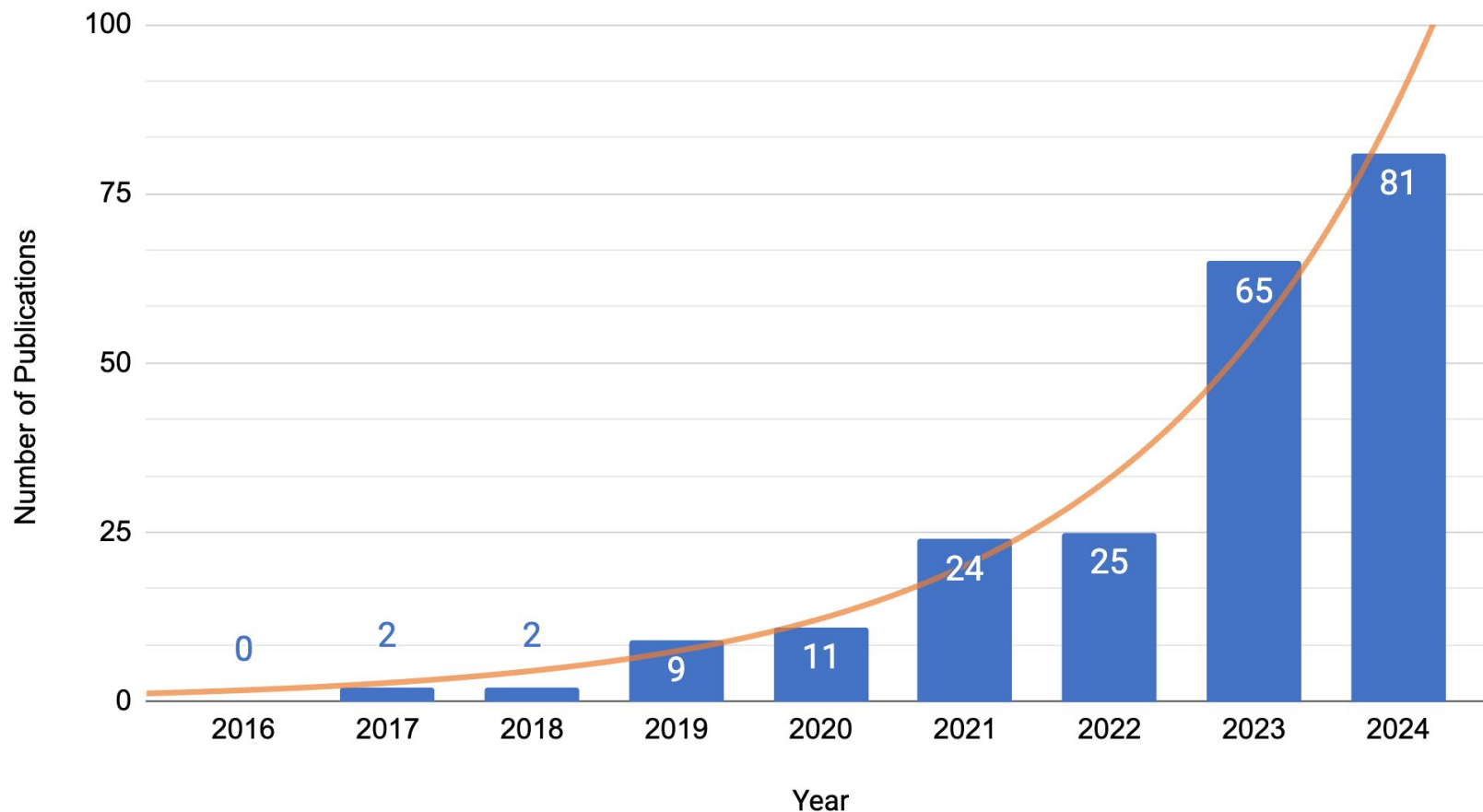
psychology literature before examining and individually coding a relevant subset of 60 articles.

The results show that chatbots are in the very beginning of entering education. Few studies suggest

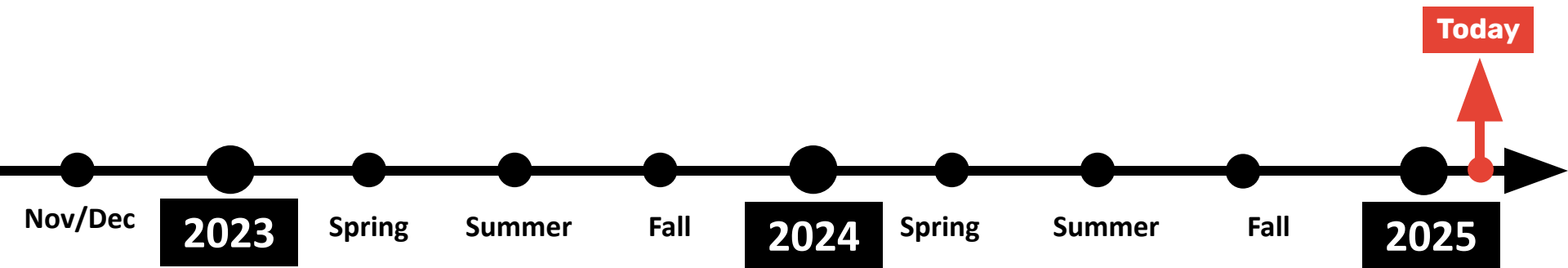
the potential of chatbots for improving learning processes and outcomes. Nevertheless, past



Occurrence of keyword "Chatbot in Education" in 2016-2024

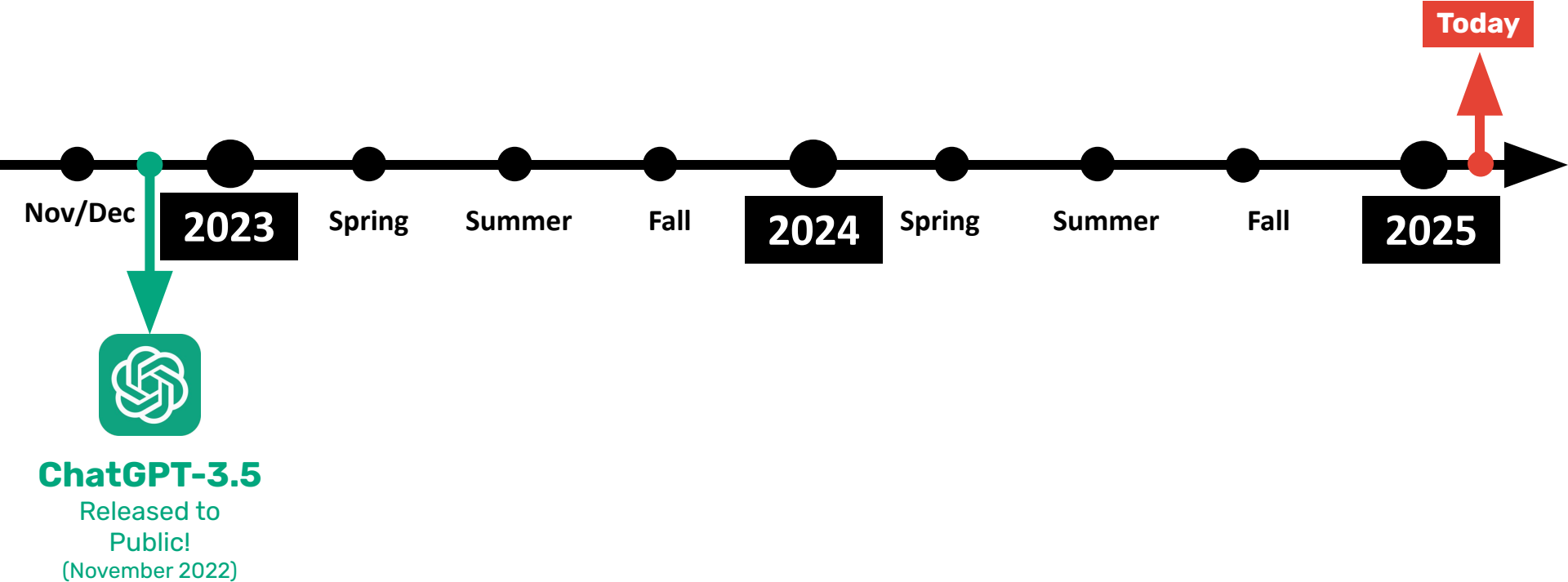


Timeline





Timeline





Timeline

GPT-4
(March 2023)



Today



Nov/Dec

2023

Spring

Summer

Fall

2024

Spring

Summer

Fall

2025

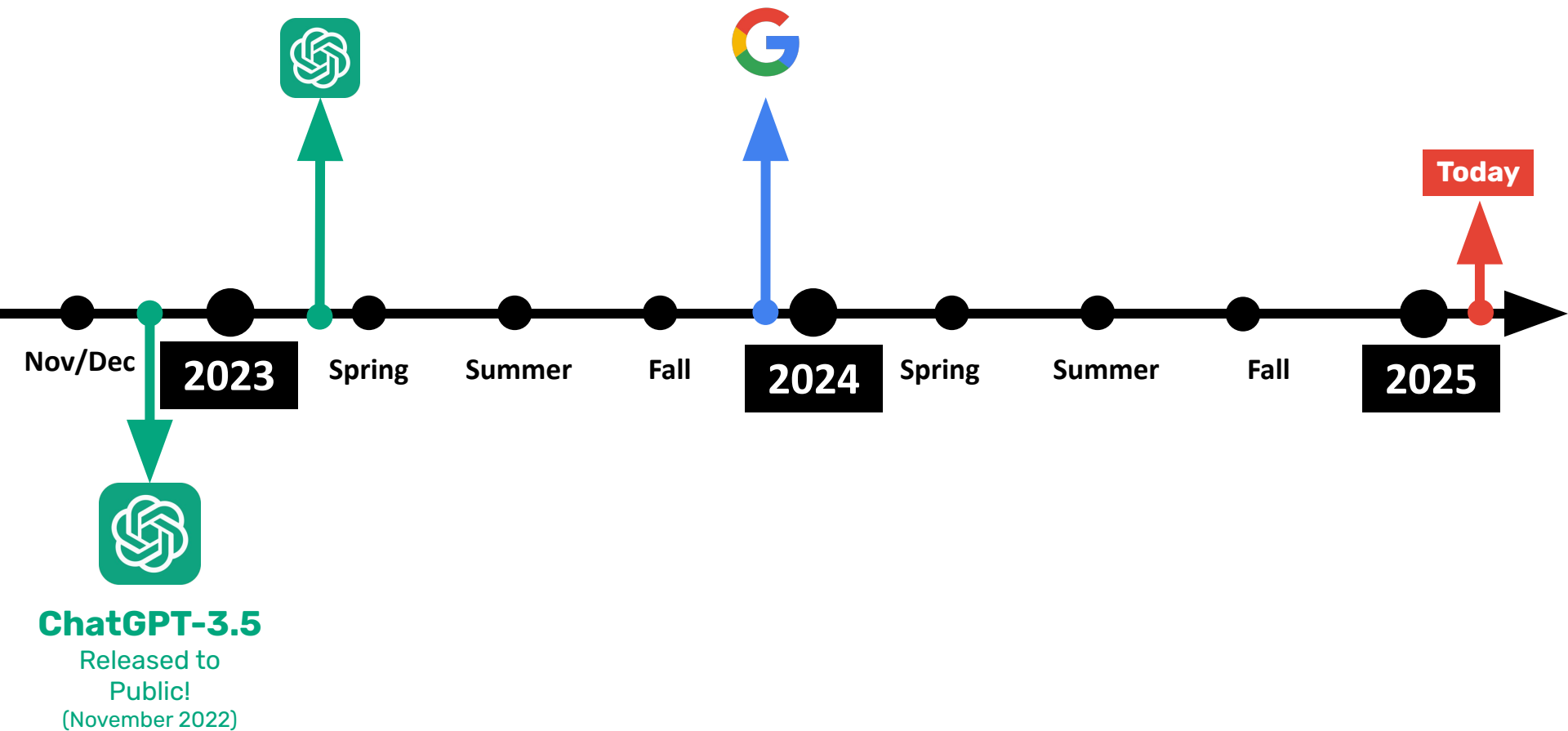


ChatGPT-3.5

Released to
Public!
(November 2022)

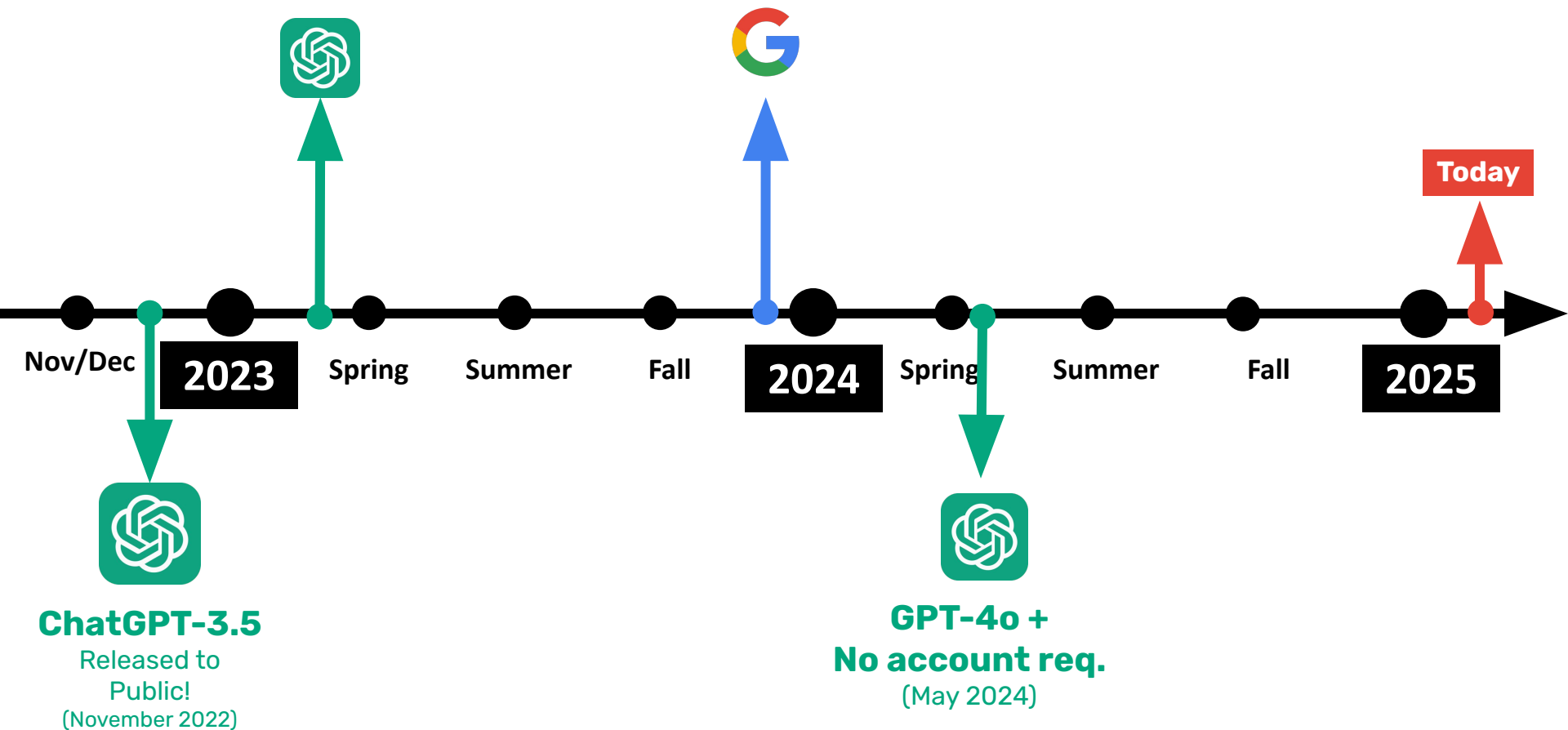


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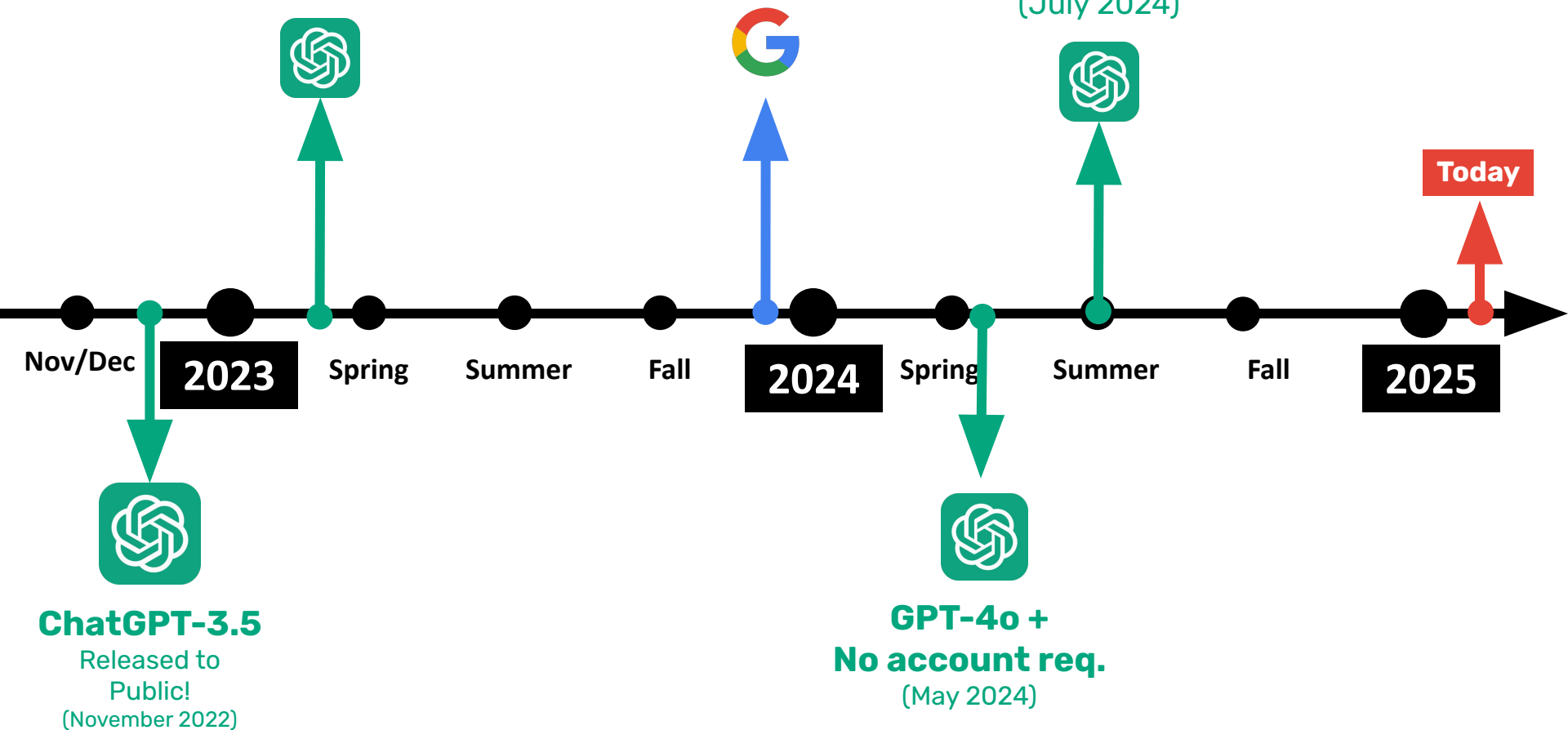


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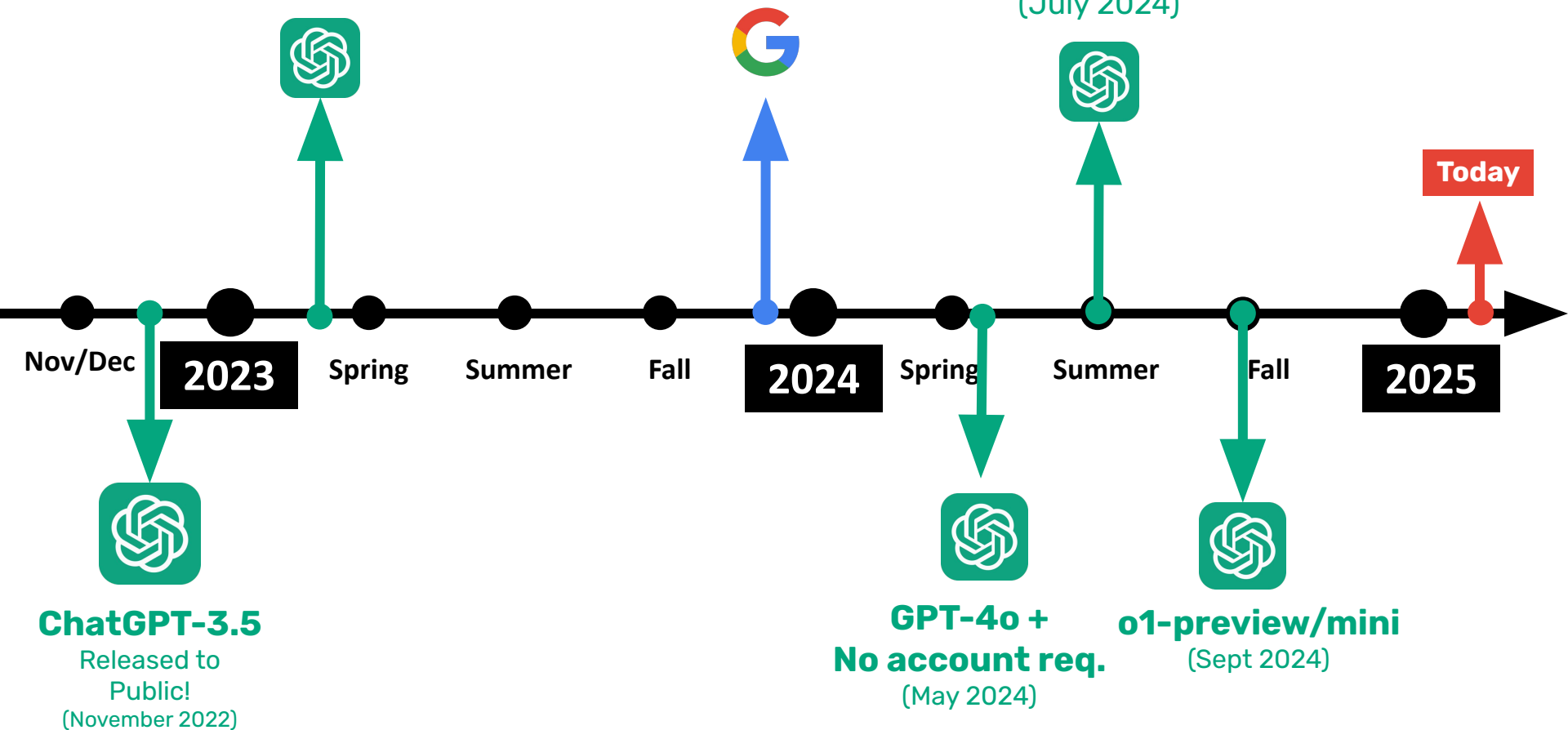


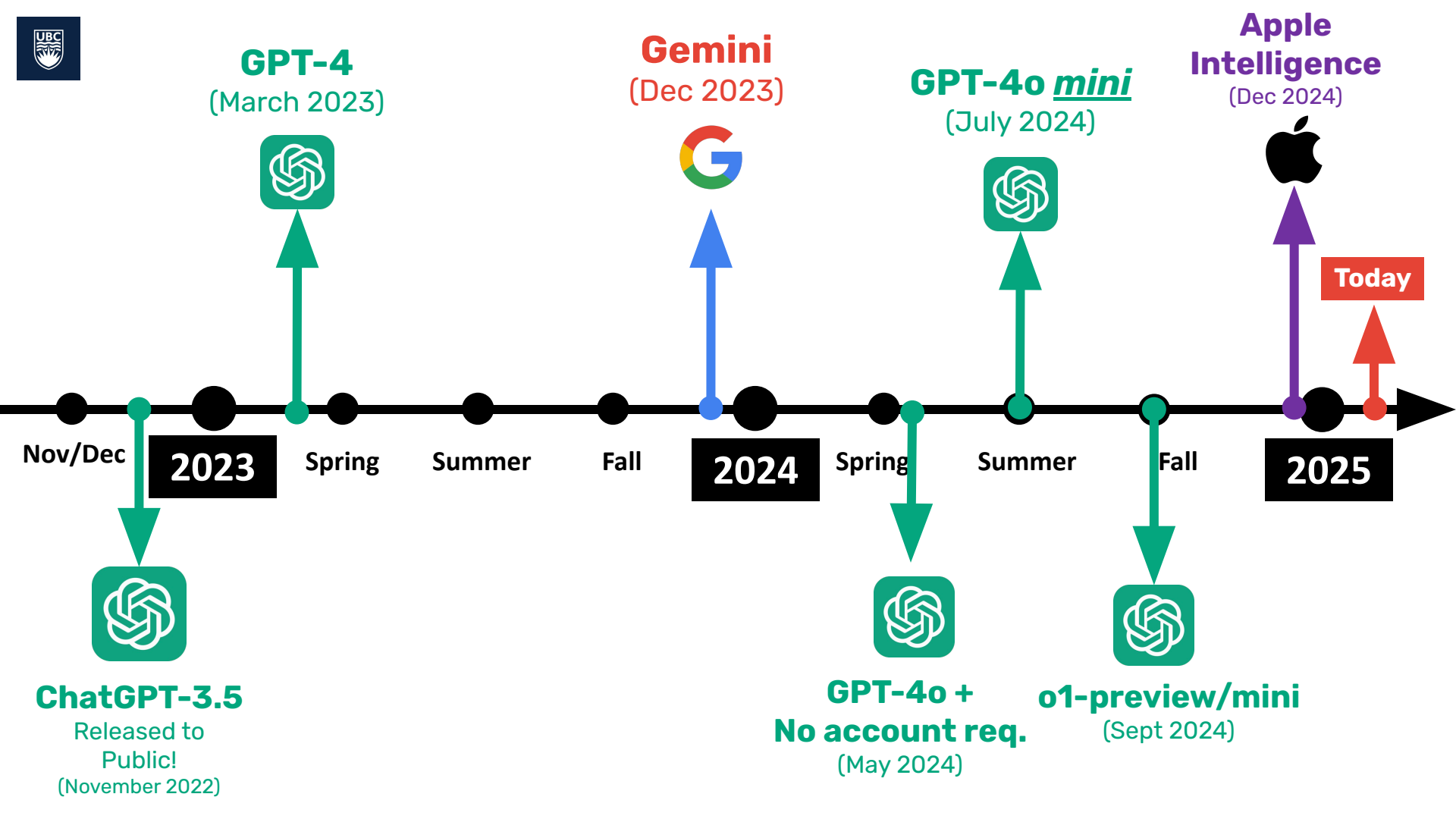
Timeline

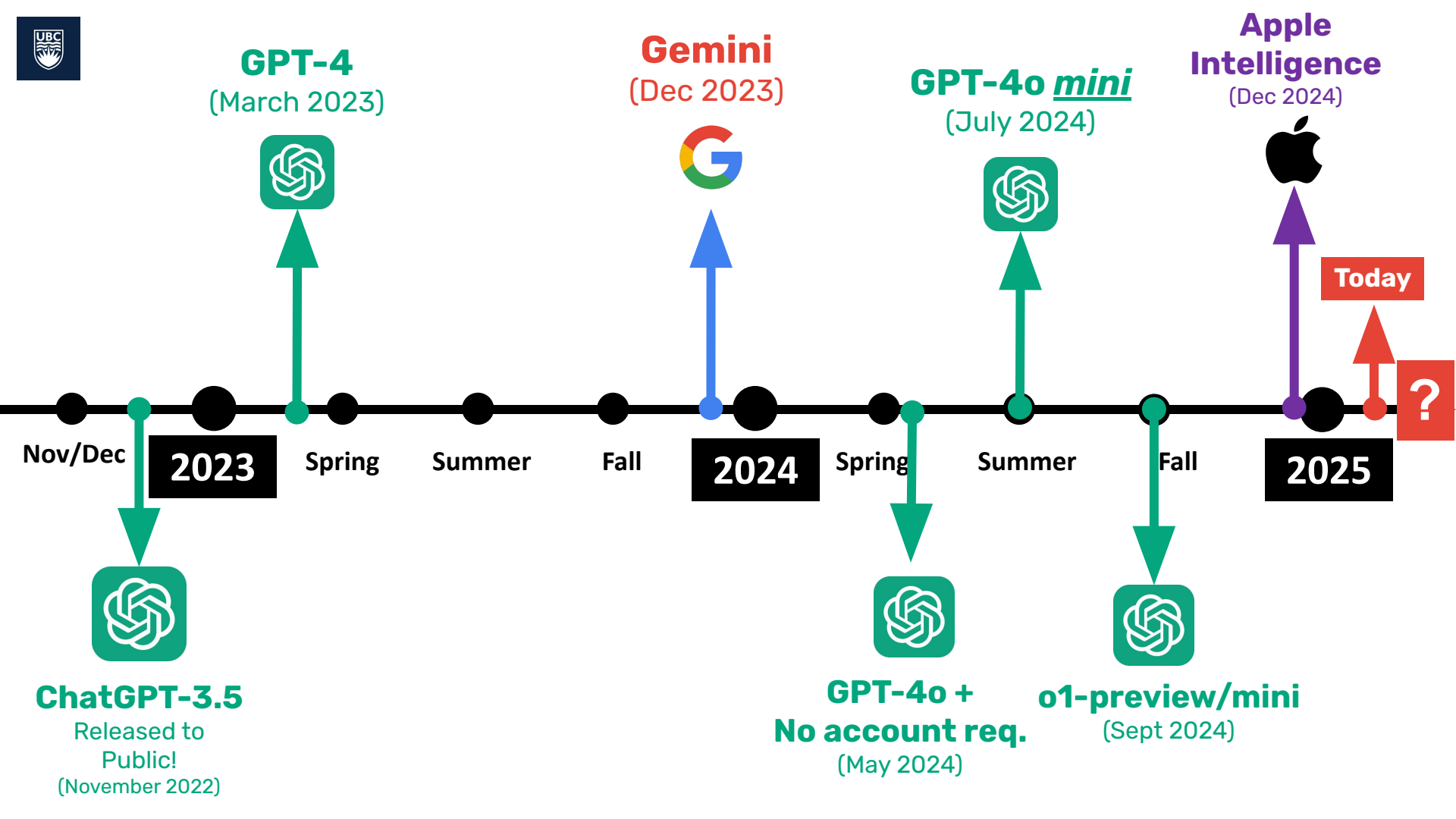




Timeline







GPT-4
(March 2023)



Gemini
(Dec 2023)



GPT-4o *mini*
(July 2024)



Apple Intelligence
(Dec 2024)



Today



Nov/Dec

2023

Spring

Summer

Fall

2024

Spring

Summer

Fall

2025

ChatGPT-3.5
Released to
Public!
(November 2022)

**GPT-4o +
No account req.**
(May 2024)

o1-preview/mini
(Sept 2024)

Linking Generative AI to Algorithms

CPSC 100 AI Policy

Class Activity



Class Activity: Course AI Policy

As a class, let's identify how AI can or can not be used in a responsible and effective way.

In groups of 2-4, brainstorm ideas on the PrairieLearn Class Activity.

Class Activities	
CA1C	Class Activity_01C
CA2A	Class Activity_02A 
CA2B	Class Activity_02B 
CA2C	Class Activity_02C 
CA3C	Class Activity_03C 





Class Activity: Course AI Policy

AI Policy

View... ▾

Develop the CPSC 100 AI Policy

In this Class Activity, you will help develop the CPSC 100 AI Policy!

Part 1

Acceptable AI use cases in CPSC 100

List some example "acceptable" AI use cases in CPSC 100

ans1.md



1



Class Activity: Course AI Policy

Part 2

Unacceptable AI use cases in CPSC 100

List some example "unacceptable" AI use cases in CPSC 100

ans2.md



1



Class Activity: Course AI Policy

Part 3

Risks of using AI in CPSC 100

Write down any risks of using AI in CPSC 100.

ans3.md



1



Class Activity: Course AI Policy

Part 4

Opportunities of using AI in CPSC 100

Write down any opportunities of using AI in CPSC 100.

ans4.md



1



Class Activity: Course AI Policy

As a class, let's identify where AI can or can not be used in a responsible and effective way.

In groups of 2-3, brainstorm ideas on the Miro Board - be ready to share!

<http://tiny.cc/100-W2A>





Class Discussion

Wrap up

Preview: Programming

This is *not* a
programming
courses

But you do need
to *understand*
how programs
work

We'll cover a small
amount of **basic concepts**
in class and you'll work on
a **visual language** in lab

From algorithms to code: **How do programs work?**

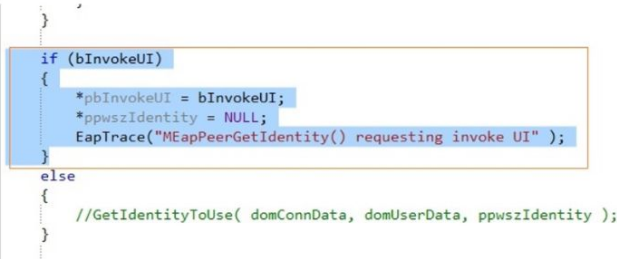
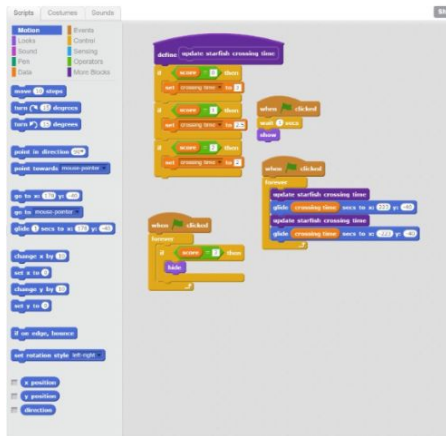
How do programs work?

Programs are a way of encoding ***algorithms*** in a precise enough way for computers to understand the instructions.

How do programs work?

Programs are a way of encoding ***algorithms*** in a precise enough way for computers to understand the instructions.

Programmers use a **high level language** like Snap, Scratch, Python, C++, Java, Racket, etc.



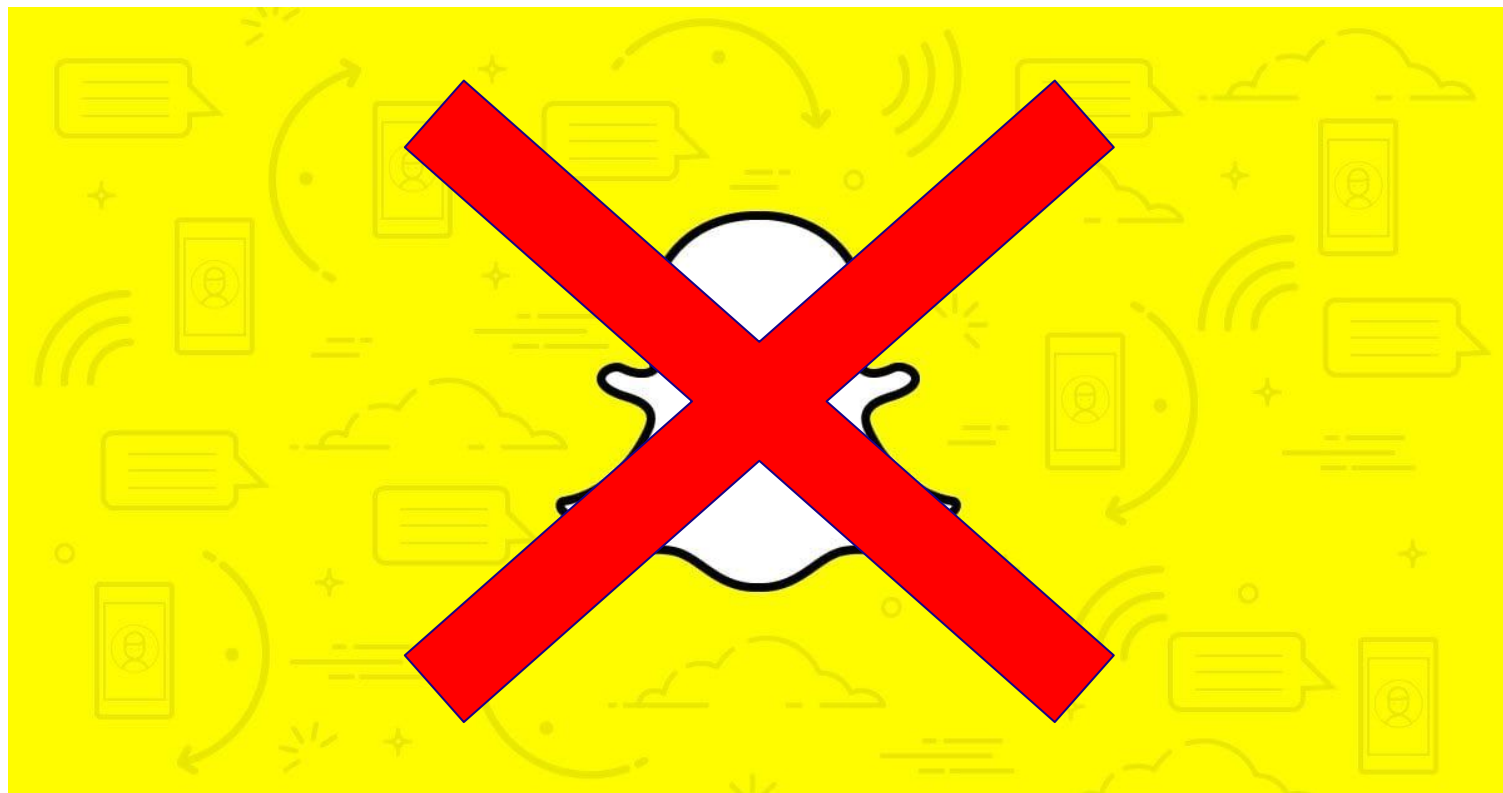
```
/**
 * Simple JButton() method.
 * @version 1.0
 * @author john doe <doe.j@example.com>
 */
HelloButton()
{
    JButton hello = new JButton( "Hello, wor
    hello.addActionListener( new HelloBtnList

    // use the JFrame type until support for t
    // new component is finished
    JFrame frame = new JFrame( "Hello Button"
    Container pane = frame.getContentPane();
    pane.add( hello );
    frame.pack();
    frame.show();           // display the fra
}
```

```
def add5(x):
    return x+5

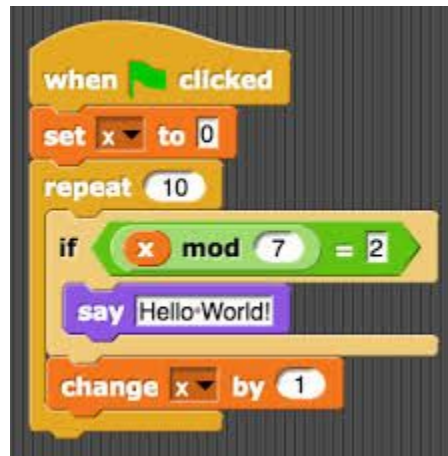
def dotwrite(ast):
    nodename = getNodeName()
    label=symbol.sym_name.get(int(ast[0]),ast[0])
    print '    %s [label="%s" % (nodename, label)
    if isinstance(ast[1], str):
        if ast[1].strip():
            print ' = %s';' % ast[1]
        else:
            print ' = ]'
    else:
        print ' = ]';'
        children = []
        for n, childrenumerate(ast[1:]):
            children.append(dotwrite(child))
        print ', ' % ast[1] -> (' % nodename
        for n, namechildren
            print '%s' % name,
```

Snap!



Demo

λ Snap!



**First, we need to
understand how
data is represented
on computers!**